original disposition matrix

transition matrix

disposition after first half-hour

disposition after first half-hour

$$\begin{bmatrix} 360 & 360 \end{bmatrix} \cdot \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \left[360 \cdot \frac{2}{3} + 360 \cdot \frac{1}{2} \;\middle|\; 360 \cdot \frac{1}{3} + 360 \cdot \frac{1}{2} \right] = \begin{bmatrix} 420 & 300 \end{bmatrix}$$